

Joel Laine

THE MICROBIOLOGICAL EFFICIENCY OF HEPA FILTRATION AND ITS USE IN MEDICAL FACILITIES

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Faculty of Medicine and Health
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Piia Sormunen
Jussi Koivumäki
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ABSTRACT

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This thesis evaluates both the microbiological efficiency of High-Efficiency Particulate Air (HEPA) filtration systems and their operational applications in medical environments. The once military-developed HEPA filtration system now serves as a vital airborne microbiological contamination control tool, since in theory it captures $\geq 99.95\%$ of $0.3\text{ }\mu\text{m}$ particles, which represent the most penetrating particle size (MPPS). The research stems from rising worldwide recognition of airborne pathogen transmission, which gained prominence during the COVID-19 pandemic. This thesis aims to act as a facilitator, helping solve the immediate need for enhanced airborne infection control in healthcare facilities.

The research begins with an overview of the different key factors in connection to airborne microbial contamination. Once a sufficient knowledge base of the underlying factors has been established, the research examines the fundamentals of HEPA filtration by studying its physical capturing mechanisms of interception, impaction, diffusion, sieving and electrostatic capture to evaluate their combined effectiveness against airborne bacteria, viruses, fungal spores and other microorganisms. The research further examines the microbiological efficiency of HEPA filtration with advanced HEPA filtration enhancements, such as antimicrobial coatings (for example silver, copper, chlorine dioxide and plant-derived substances), integration of nanofibre materials and combined technologies (for example ultraviolet irradiation, photocatalytic oxidation and air ions) used to improve microbial capture and inactivation. In addition, the research investigates how environmental factors including humidity levels, airflow patterns, ventilation systems and maintenance procedures affect filtration performance and microbial inactivation. To address the regulatory aspects, this thesis examines HEPA filtration guidelines, standards and regulations in Finnish healthcare facilities, while comparing them to German standards.

Based on the research, recommendations for the use of HEPA filtration in medical facilities are derived and presented as part of the conclusions. In conclusion, the research demonstrates that HEPA filters are proficient and, in many cases, excel at removing diverse microbial contaminants, but their microbiological effectiveness depends on countless factors including the environment, ventilation rate, proper filter placement, pre-filtration and strict maintenance procedures to stop microbial growth on filter media. Based on the research, the estimated microbiological efficiency of unmodified HEPA filters is $\sim 70\text{--}100\%$ for fungi and $\sim 60\text{--}100\%$ for bacteria. A justifiable estimate for the viral efficiency could not be formulated due to prevailing conflicting research.

Keywords: HEPA filtration, indoor air quality, microbes, microbiological efficiency, airborne pathogens, medical facilities, ventilation, antimicrobial technologies, standards and regulations

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TIIVISTELMÄ

Joel Laine : HEPA filttereiden mikrobiologinen tehokkuus ja käyttö lääketieteellisissä tiloissa
Kandidaatintutkielma
Tampereen yliopisto
Bioteknologian ja biolääketieteen tekniikan koulutus
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Tämä kandidaatintyö arvioi sekä HEPA-suodatuksen (engl. High-Efficiency Particulate Air) mikrobiologista tehokkuutta että sen toiminnallisia sovelluksia lääketieteellisissä ympäristöissä. Alun perin sotilaalliseen käyttöön kehitetty HEPA-suodatusjärjestelmä toimii nyt erittäin tärkeänä ilmateitse leviävän mikrobiologisen kontaminaation hallintamenetelmänä, sillä teoriassa se vangitsee $\geq 99,95\%$:a $0,3\text{ }\mu\text{m}$:n hiukkasista, jotka edustavat tunkeutuvinta hiukkaskokoa (MPPS). Tutkimuksen taustalla on kasvava maailmanlaajuinen tietoisuus ilmateitse tapahtuvasta taudinaiheuttajien leviämisestä, joka tuli erityisen ajankohtaiseksi COVID-19-pandemian aikana. Tämä kandidaatintyö pyrkii toimimaan edistäjänä ja osaltaan vastaamaan välittömään tarpeeseen tehostaa ilmateitse leviävien infektioiden torjuntaa terveydenhuollon laitoksissa.

Tutkimus alkaa katsauksella erilaisiin avaintekijöihin, jotka liittyvät ilmateitse tapahtuvaan mikrobikontaminaatioon. Kun riittävä tietopohja taustatekijöistä on luotu, tutkimus tarkastelee HEPA-suodatuksen perusteita tutkimalla sen fysikaalisia hiukkasten vangitsemismekanismeja: sieppaus, törmäys, diffuusio, seulonta ja sähköstaattinen vangitseminen, arvioidakseen näiden yhteisvaikutuksen tehokkuutta ilmateitse leviävien bakteerien, virusten, sienimikro-organismien ja muiden mikro-organismien poistamisessa. Tutkimus tarkastelee lisäksi HEPA-suodatuksen mikrobiologista tehokkuutta edistyneiden HEPA-suodattimien parannusten avulla, kuten antimikrobisten pinnoitteiden (esimerkiksi hopea, kupari, klooridioksidi ja kasvipohjaiset aineet), nanokuitumateriaalien integroinnin sekä yhdistettävien teknologioiden (esimerkiksi ultraviolettisäteilytys, fotokatalyyttinen hapetus ja ilman ionit) avulla, jotka parantavat mikrobien sieppausta ja inaktivointia. Lisäksi tutkimus selvittää, kuinka ympäristötekijät, mukaan lukien kosteustasot, ilmavirtaukset, ilmanvaihtojärjestelmät ja huoltokäytännöt vaikuttavat suodatuksen suorituskykyyn ja mikrobien inaktivointiin. Sääntelynäkökohtien huomioimiseksi tutkielmassa tarkastellaan HEPA-suodatuksen ohjeistuksia, standardeja ja määräyksiä suomalaisissa terveydenhuollon laitoksissa, vertaillen niitä Saksan standardeihin.

Tutkimuksen perusteella johdetaan ja esitetään suosituksia HEPA-suodatuksen käytöstä lääketieteellisissä laitoksissa osana johtopäätöksiä. Yhteenvedona tutkimus osoittaa, että HEPA-suodattimet ovat tehokkaita ja monissa tapauksissa erittäin tehokkaita poistamaan erilaisia mikrobikontaminantteja, mutta niiden mikrobiologinen tehokkuus riippuu lukuisista tekijöistä, kuten ympäristöstä, ilmanvaihdon nopeudesta, suodattimen oikeasta sijoittelusta, esisuodatksesta ja tiukoista huoltokäytännöistä mikrobikasvun estämiseksi suodatinmateriaalissa. Tutkimuksen perusteella arvioitu mikrobiologinen tehokkuus muokkaamattomilla HEPA-suodattimilla on $\sim 70\text{--}100\%$ sienimikro-organismeille ja $\sim 60\text{--}100\%$ bakteereille. Virusten osalta perusteltua tehokkuusarviota ei voitu muodostaa vallitsevan ristiriitaisen tutkimustiedon vuoksi.

Avainsanat: HEPA-suodatus, sisäilman laatu, mikrobi, mikrobiologinen tehokkuus, ilmateitse leviävät taudinaiheuttajat, terveydenhuollon laitokset, ilmanvaihto, antimikrobiset teknologiat, standardit ja määräykset

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck –ohjelmalla.

PREFACE

This thesis was written as part of my bachelor's studies in Biotechnology and Medical Technology at Tampere University. The work focuses on evaluating the microbiological efficiency of HEPA filtration systems and their operational applications in medical facilities.

I would like to thank my supervisor, Piia Sormunen, for her guidance, feedback and support throughout the entire research process. I would also like to thank Jussi Koivumäki for orchestrating this thesis course well and for offering his support when needed. Furthermore, I wish to express my gratitude to everyone who supported me during the process, especially my family.

In Tampere, Finland, on 31 May 2025

Joel Laine

THE USE OF AI

Artificial intelligence (AI) was used in the making of this thesis:

- ☐ No
- ☒ Yes

The AI tools used in my thesis and the purpose of their use has been described below:

Semantic Scholar: An academic search engine that uses AI to find relevant scholarly papers.

Use: Semantic Scholar was used to find relevant research, but only in small quantities, as most of the research had already been gathered through Andor and Google Scholar prior to the use of Semantic Scholar.

ChatGPT-4o: OpenAI's advanced AI model that can process and generate text, images and audio.

Uses: ChatGPT-4o was used in brainstorming ideas, understanding concepts (mainly the regulatory structure), making sure that all the key components regarding broader concepts were taken into account, finding better suiting synonyms for words and reorganizing/rewording sentences. Also, a translation of the Abstract was done into Finnish, which was then further edited. This was done to preserve a structure as similar as possible between the Abstract and its Finnish counterpart. To confirm, the analytical content and argumentation were produced by myself. ChatGPT-4o was used solely as a language and structural support tool, without outsourcing original thinking or academic analysis.

I am aware that I am fully responsible for the entire content of this thesis, including the parts generated by AI, and accept the responsibility for any violations of the ethical standards of publications.

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LIST OF ABBREVIATIONS

ACH	Air Changes per Hour
AI	Artificial Intelligence
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BAuA	German Federal Institute for Occupational Safety and Health
BfArM	German Federal Institute for Drugs and Medical Devices
BfR	German Federal Institute for Risk Assessment
CDC	Centre for Disease Control and Prevention
CEN	European Committee for Standardization
CFU	Colony-Forming Unit
COVID-19	Coronavirus Disease 2019
DGKH	German Society for Hospital Hygiene
DGUV	German Statutory Accident Insurance
DIN	German Institute for Standardization
DNA	Deoxyribonucleic Acid
DOP	Dioctyl Phthalate
ECDC	European Centre for Disease Prevention and Control
EN	European Norm
EPA	Efficient Particulate Air
EU	European Union
FISE	Finnish Qualification Requirements in the Building Sector
HEPA	High-Efficiency Particulate Air
HVAC	Heating, Ventilation and Air-Conditioning
IAQ	Indoor Air Quality
IfSG	German Infection Protection Act
ISO	International Organization for Standardization
MPPS	Most Penetrating Particle Size
PCO	Photocatalytic Oxidation
PE	Polyethylene
PM	Particulate Matter
PP	Polypropylene
PS	Polysulfone
PTFE	Polytetrafluoroethylene
PVDF	Polyvinylidene Fluoride
RH	Relative Humidity
RKI	Robert Koch Institute
RNA	Ribonucleic Acid
SARS-CoV	Severe Acute Respiratory Syndrome Coronavirus
SFS	Finnish Standards Association
THL	Finnish Institute for Health and Welfare
Tukes	Finnish Safety and Chemicals Agency
ULPA	Ultra-Low Penetration Air filter
UVGI	Ultraviolet Germicidal Irradiation
VALVIRA	Finnish National Supervisory Authority for Welfare and Health
VDI	Association of German Engineers
VOC	Volatile Organic Compound
WHO	World Health Organization
ZLS	German Central Office of the Federal States for Safety Technology

1. INTRODUCTION

1.1 Background and Significance

COVID-19 (Coronavirus Disease 2019), more commonly known in everyday language as the coronavirus, took the world by surprise in 2019, as its name suggests. According to the World Health Organization (WHO) approximately five years later, in September 2024, over 776 million cases had been reported worldwide, with more than 7 million of those resulting in the patient's death (Paasi 2024). The COVID-19 pandemic drastically reshaped our understanding of airborne infection risks in indoor spaces. Hospitals, dental clinics, waiting rooms and even break areas were recognized not just as treatment spaces, but also as potential hubs of transmission. In general, it is believed that around 69–80% of all pathogens that cause infections are airborne (Urrutia et al. 2023). Thus, this global event intensified awareness of how easily microbes can travel through the air and emphasized the need for effective airborne infection control strategies in medical facilities (WHO 2021).

By 2050, some 70% of humanity will live in urban areas. It is also believed that humans spend ~70–80% of their time indoors (Goldstein et al. 2021). Increasing urbanization, rapid technological development and an increase in specialized professions has given rise to an 'indoor generation'—a generation that spends more of its time in enclosed indoor environments, characterized by complex and chemically diverse air compositions (Kelly & Fussell 2019).

High-Efficiency Particulate Air (HEPA) filters have been widely adopted in healthcare settings as a non-pharmaceutical method to reduce exposure to airborne pathogens. In theory, HEPA filters are capable of removing $\geq 99.95\%$ of airborne particles at the most penetrating particle size (MPPS) and are even more efficient with both smaller and larger particles due to combined capture mechanisms such as diffusion, interception and impaction (Camfil 2021; Cheremisinoff 2016). While the basic function of HEPA filters—to trap airborne particles—is well established, their true microbiological efficiency depends on much more than the filter's removal capacity alone. Environmental factors such as humidity, temperature, airflow and system design, as well as filter placement, maintenance routines and the use of complementary technologies all influence the outcome.

Public health institutions and governments have begun to adopt a more proactive stance toward indoor air quality (IAQ). In Finland, for instance, HEPA-based air purification was used as part of the E3 pandemic response in kindergartens, reducing children's sick days by up to 30% (Terävä 2023). In addition to other health benefits, removing common contaminants from indoor air has been observed to have a considerable positive impact on, for example, the productivity of humans (Kelly & Fussell 2019). Yet, despite their proven utility, the use of HEPA filters in medical facilities is not uniformly regulated by law across Europe.

This thesis investigates the microbiological efficiency of HEPA filtration in medical facilities, focusing not only on the filter technology itself but also on the broader system-level and environmental factors that influence its performance. The goal is to provide evidence-based insights and practical recommendations for improving airborne infection control in healthcare environments.

1.2 Objectives

This study will provide the reader with a comprehensive understanding of HEPA air filtration and its underlying influencing factors, with an emphasis on their microbiological efficiency and use in medical facilities. This study will act as a facilitator for the improvement of IAQ in medical facilities.

Research questions:

- 1) How effectively do HEPA filters remove microbiological contaminants such as bacteria, viruses and fungal spores, and how can filter performance vary?
- 2) What are the capabilities and limitations of HEPA filtration in preventing microbial contamination in medical facilities? How can the capabilities be enhanced and the limitations be reduced?
- 3) What entities or authorities govern and regulate the use of HEPA filters in medical facilities in Finland? How do these compare to their counterparts in Germany, which is known for its strict regulatory framework?

Based on this study, general recommendations for the use of HEPA filtering in medical facilities will be derived, which will be presented as part of the conclusions. These recommendations can later be verified and improved upon with a future study.

1.3 Research Plan and Scope

This thesis will include viruses as microbiological life, since they are extremely prevalent in medical facilities. This thesis limits its scope to HEPA filtration systems used in indoor medical environments, both integrated into heating, ventilation and air-conditioning (HVAC) systems and as portable air purifiers. The primary objective is to evaluate microbiological filtration efficiency through experimental studies, peer-reviewed literature, technical standards and expert guidelines. The research also seeks to analyse how ventilation design, maintenance routines, environmental conditions and supplementary technologies impact filter effectiveness and microbe inactivation. Ultimately, the goal is to generate evidence-based recommendations for the deployment of HEPA filtration in hospitals and similar facilities to reduce airborne transmission risks and improve indoor environmental quality more broadly.

All information used in this thesis will be collected from credible and verifiable sources. Scholarly literature will be prioritized whenever possible to ensure academic reliability and all references will be cited according to fair use and compiled in a comprehensive reference list at the end of the thesis. While some sources will be older than others, their information remains highly relevant, as the fundamental principles of HEPA filtration have remained largely unchanged over time.

Primary academic sources will be retrieved from platforms such as Andor, Google Scholar and Semantic Scholar. Additional supporting materials—such as regulatory documents, technical guidelines, or industry publications—will be gathered through reputable search engines, primarily Google. The aim is to incorporate as many trustworthy sources as possible to construct a comprehensive and well-informed understanding of the topic.

2. AEROBIOLOGY

Aerobiology refers to the study of ubiquitous biological particles in ambient air, both indoors and outdoors. Particles of biological origin, which can be suspended in air for variable periods of time due to their small sizes, “can include viruses, bacteria, actinomycetes, fragments of fungi and fungal spores, lichen fragments and their spores, protists, spores of plants, pollen, plant fragments and small seeds, invertebrates and their fragments and faecal material, plus skin, hair, dried mucus and excrement from larger animals” (Lacey & West 2006).

2.1 Particle dynamics

Particulate matter (PM) is an all-encompassing term used to group all categories of small particles. Usually, PM is divided into PM₁₀ (particles under 10 µm in diameter) and PM_{2.5} (particles under 2.5 µm in diameter), of which the latter is considered to have more adverse effects on health due to their higher penetration ability. For example, pollen, mould spores, fine ash and tobacco smoke are 9–80 µm, 3–50 µm, 0.7–60 µm and 0.1–7 µm in size, respectively (Elsaid et al. 2021).

The settling velocity varies according to the size and mass of the particle. In general, particles with a diameter of under 1 µm, 10 µm and 100 µm have uninterrupted settling velocities under 0.01 cm/s, 1 cm/s and 100 cm/s, respectively (Nazaroff 2022). This means that in a space with no external airflows, a 1 µm particle can take over 10 h to fall from a human mouth's height, while 100 µm particles take around 5 s (Wang et al. 2021). In general, the smaller the particle, the longer it takes to settle. Airflows have the potential to prolong or abbreviate these times drastically (Tellier et al. 2019). The smaller the particle, the more the particle tends to migrate (Mousavi et al. 2020). Smaller particles are also much more easily resuspended.

Pollutants of notable concern that infiltrate indoors include PM, nitrogen dioxide (NO₂) and ozone (O₃) to name a few (Kelly & Fussell 2019). PM concentrations outdoors can vary substantially due to, for example, sandstorms, hazy days and other air/weather related phenomena (Zhai et al. 2018). Factors governing pollutants indoors can be categorized based on the attributes of the pollutants (chemical structure, dynamic properties, outdoor air concentration et cetera), building in question (design, construction, state of repair et cetera) and the timing of presence (occupant density, lifestyle, habits, behaviours et cetera) (Kelly & Fussell 2019). The general estimated infiltration factor, which is

a measurement that avoids the influence of indoor sources of PM is somewhere around 0.5 for $PM_{2.5}$ (Goldstein et al. 2021), but it depends on countless factors. In Helsinki, the estimated infiltration factor for $PM_{2.5}$ is around the average, while the estimated infiltration factor for PM_{10} is around 0.2 (Chen & Zhao 2011). Humans spend most of their time indoors, where finer PM is more likely to infiltrate, hence we tend to inhale most of the finer outdoor PM inside (Goldstein et al. 2021). Particles with a diameter of 0.1–1 μm have a higher infiltration factor of 0.8–1, while the infiltration factors of particles smaller and larger than that quickly plummet close to 0 (Chen & Zhao 2011), which means that particles with a diameter of 0.1–1 μm have a higher chance to penetrate a building compared to other sizes. This is due to the dynamics of particles $>1 \mu m$ and $<0.1 \mu m$; larger ones have a higher gravitational settling velocity and inertial impaction rate, while smaller ones are more affected by Brownian motion (Nazaroff 2016). The concentration of gas-phase organic compounds is usually slightly higher indoors. When outdoor $PM_{2.5}$ travels indoors, it sorbs semi-volatile organic compounds. These compounds can also infiltrate indoors, but more likely are released from indoor sources. This combination leads to an even more toxic PM with the same ability to penetrate deep into the respiratory system (Goldstein et al. 2021).

The I/O ratio refers to the ratio of indoor to outdoor particle concentration. The $PM_{2.5}$ levels indoors are generally lower than outdoors, but in densely populated areas, such as cities, the I/O ratio can be over 3 (Chen & Zhao 2011). This is because PM accumulates indoors. In heavily populated areas there is much more PM outside, leading to a larger accumulation indoors. In Helsinki, the I/O ratio is estimated to be around 1 for $PM_{2.5}$, but the ratio changes for example according to the weather and season, similarly to other air pollutants (Chen & Zhao 2011; Ghosh et al. 2015; Wan et al. 2011). $PM_{2.5}$, PM_{10} and total volatile organic compounds (VOC) typically increase during warm seasons (Cham-seddine et al. 2019), but it is not a straight indication of a change in the I/O ratio (Rivas et al. 2015). Seasons can influence the I/O ratios of different PM in different ways. Older buildings tend to have more cracks in their outer shell, possibly leading to a higher I/O ratio (Rivas et al. 2015). PM can also penetrate indoors for example via ventilation (Chen & Zhao 2011). The most common symptoms of poor IAQ include headaches, irritation in the eyes, throat and nose, fatigue, dry mucous membranes and skin, hypersensitivity, nausea, colds and dizziness (Kumar et al. 2022).

In general, the most important factor in keeping different PM levels in buildings as low as possible is proper ventilation. As humans move inside buildings, the indoor air gets contaminated with PM. Since PM takes a long time to settle, it can stay suspended in the air for long periods of time even in stagnant zones, where there is minimal airflow

(Tsang et al. 2023). Ventilation aims to minimize the concentration of contaminants in indoor air by removing as many contaminants as possible, while supplying fresher, less contaminated air. There are numerous different ventilation strategies and system types, and the concentration of indoor bioaerosols has been observed to positively correlate with the type of ventilation system in use (Ting Wu et al. 2021). At least sophisticated mechanical ventilation and hybrid ventilation usually involve a filtration component. However, the accumulation of dust and moisture in unmaintained ventilation systems have been observed to increase the risk of transmission of microbes (Ting Wu et al. 2021; Stockwell et al. 2019; Ribeiro et al. 2024). Some ventilation systems, especially mechanical ones, can also create or continue to resuspend bioaerosols (Stockwell et al. 2019; Izadyar & Miller 2022). This is why proper planning, cleaning and maintenance is so crucial for these systems to have a lasting positive effect on air quality.

The most important overall contaminant decreasing factor in ventilation is the ventilation rate, since lower infection probabilities can be achieved with higher ventilation rates alone (Dai & Zhao 2020). Simply put, a higher ventilation rate means that the air is changed more often, which leads to a lower concentration of PM in indoor air (Nazaroff 2016; Elsaid et al. 2021; Kelly & Fussell 2019; Izadyar & Miller 2022; Tsang et al. 2023; Pasquarella et al. 2000). Ventilation rate is often measured in air changes per hour (ACH), which measures the total times the entire volume of air in a given space is replaced in one hour. In one study, infection rates were cut by half with an increase in ACH from 2 to 6 (Srikrishna 2022). For example, operating rooms in hospitals have a high recommended ACH of around 20 (Wan et al. 2011).

2.2 Micro-organisms in Indoor Air

The air spora is the population of airborne particles of plant or animal origin; pollen, plant and fungal spores, protists (protozoa), bacteria, viruses and fragments of any biological origin. Since the particles are incapable to stay airborne on their own, naturally they descend due to gravity (Lacey & West 2006). The air spora shows a wide variety in particle sizes; all sizes small enough where the effects of gravity are reduced and the particle can stay suspended in the air for longer. Smaller airborne micro-organisms are often adhered to other PM, but they can also exist individually (Zhai et al. 2018). A biological particle with the mass of a few hundred microns, depending on the surroundings, can stay suspended in the air for a period of time long enough for it to be considered aerobiology (Lacey & West 2006). Particles larger than a few hundred microns start to have much more gravitational influence, leading to an exponentially faster descent (Nazaroff 2022).

There are endless ways to categorize particle sizes, but to give some perspective, particles over 100 μm in diameter are considered very large and settle rapidly, and particles under 1 μm in diameter are considered very small and settle slowly. They are heavily impacted by airflows around them, which can carry smaller particles for longer distances. Microbial airborne PM has been observed to behave similarly to abiotic PM of the same aerodynamic size, which is supported by both theoretical and experimental evidence (Nazaroff 2016). Microbial contaminants in the air can be divided into viable and non-viable biological pollutants (Liu et al. 2018).

The air spora consists of harmless and harmful PM, although it can be debated that particles in general cause more harm than good when inhaled. PM can cause a wide variety of diseases; cardiovascular problems, ocular problems, neurological problems, communicable respiratory diseases et cetera (Kumar et al. 2023). Harmful particles may cause diseases in humans and animals through mechanisms such as direct infection, toxicosis, or allergic responses (Lacey & West 2006; Shiraiwa et al. 2017). Some possible harmful mechanisms include oxidative stress, deoxyribonucleic acid (DNA) methylation, epigenetic modifications and gene activation (Kumar et al. 2023). The effects on a specific subject's health can vary depending on the individual; effects on the subject can range from none to fatal. For example, some harmful particles can be symptomless or just slightly irritate some part of the subject's outer layer, while another particle might penetrate deep into the subject and cause a genetic mutation in its genome, leading to the development of cancer and ultimately death. Particles can also affect different subjects in different ways.

Humans are major sources for biological particles in indoor air. In addition to biological particles, humans emit many other produces directly and indirectly, for example gasses, abiotic particles and inorganic compounds, of which many can be harmful especially in larger quantities. For example, several hundred VOCs are known to be emitted by humans into their surroundings (Goldstein et al. 2021). PM adheres to humans when we are outside, which we then carry inside and vice versa (Layton & Beamer 2009). It is estimated that in a seated condition a human emits approximately 10^6 particles per hour into the ambient air, which adds up to ~ 30 mg per person per hour. However, this can vary a fair amount depending on the individual. In the same timeframe, humans have been found to emit around 3.7×10^7 and 7.3×10^6 bacterial and fungal genome copies, respectively (Prussin & Marr 2015). These particles have a high likelihood to get airborne, for example, in the process of shedding skin, either attached to the shedded skin cell or unattached. It has been observed that microbes found in a house are distinctively related to the occupants, and when the occupants change, so does the microbial activity.

This correlates with every human having their own microbiota (Nazaroff 2016; Lax et al. 2014). For example, bacteria are liberated from humans into the ambient air at a rate of about 1000 per minute during just slight movement, while this amount increases by around 10-fold when the activity level is raised to marching (Nazaroff 2016).

D_{50} , the mean dose that causes an infection in 50% of susceptible subjects, varies greatly depending on the microbe in question. Different studies have estimated the D_{50} being around 790 viral copies for Influenza-A, 280 viral copies for SARS-CoV-1 (Severe Acute Respiratory Syndrome Coronavirus 1) (around the same for common cold and animal coronaviruses) and <1000 viral copies for SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2) (Lelieveld et al. 2020).

Bacterial cells, viruses and spores all depend on external transportation when travelling for longer distances. They come in a wide variety of shapes and relative sizes. When broadly classifying their sizes, viruses, bacteria and fungi are typically around 0.02–0.2 μm , 0.2–5 μm and 0.5–30 μm (with exceptions), respectively (Azimi & Stephens 2013; Stålfelt et al. 2023; Morawska 2006; Zhai et al. 2018). Due to their small sizes, airborne transportation is a very viable option. The levels of bacteria, viruses and fungi correlate with human activity; levels tend to be higher in buildings with a higher indoor activity level (Nazaroff 2016). In general, most of the airborne micro-organisms are bacteria (usually >50%), with fungi as a close second (around 0.5–1 times the concentration of bacteria). Viruses typically have the smallest concentrations. Usually, bacteria make up around 90% and 80% of micro-organisms in $\text{PM}_{2.5}$ and PM_{10} , respectively, and the rest consists mostly of fungi. Indoor levels of bacteria and fungi are often much closer to each other, while outdoors the fungi can overtake by many folds (Zhai et al. 2018). Different types of micro-organisms favour different temperatures and humidity ranges (Liu et al. 2018).

Bacteria can be suspended in the air alone, clustered with other bacteria or adhered to other PM (Nazaroff 2016; Guo et al. 2020). They tend to adhere the most to particles sized 4–20 μm (Stålfelt et al. 2023). It is estimated that 10–20% of endemic nosocomial infections are caused by airborne bacteria (Mirhoseini et al. 2015). In occupied spaces, bacterial levels tend to be much higher than in the outdoor air. Across all particle sizes, the geometric mean level of endotoxin, which is a substance associated with gram-negative bacteria, was around 50% higher indoors compared to outdoors (Nazaroff 2016). The viability of bacteria declines if the temperature is above 24°C (Liu et al. 2018). Increases in bacterial or fungal loads in the air have been observed to correlate with decreases in maximum breathing capacity in asthmatic patients (Zemouri et al. 2017).

Fungal levels depend on the environment, but generally airborne fungal levels are lower than bacterial levels indoors (Mirhoseini et al. 2016). Fungi grow actively within roughly 15–40°C and their growth and reproduction generally accelerate in a relative humidity (RH) range of 40–90% (Liu et al. 2018). Fungi concentrations are frequently higher in outdoor than indoor air. The largest published survey of indoor bioaerosol levels, carried out in the United States, reported a geometric mean outdoor and indoor concentration of 493 CFU/m³ (colony-forming units per meter squared) and 79 CFU/m³ for fungi, respectively. This means that fungi concentrations are commonly higher in outdoor air. Glucan levels, which are a marker of fungi, are much higher in particles larger than 10 µm in diameter, which indicates that most of the fungi matter is larger than 10 µm (Nazaroff 2016). This information correlates well with the average size of fungi and the lower infiltration factor for larger particles partly explains the differences between outdoor and indoor geometric means.

The lifetime of viruses in the air is a function of physical transport and biological inactivation, both of which are modulated by environmental conditions such as humidity, temperature and ultraviolet radiation (Wang et al. 2021). Most of the viral load tends to be in particles smaller than 2.5 µm (Nazaroff 2016). Some viruses survive better in higher RH, while others prefer a lower RH. The minimum infectious dose has been observed to vary based on virus type and deposition site within the respiratory tract; particles that deposit deeper in the respiratory tract could require less virus to initiate infection (Wang et al. 2021). For example, about 64% of coronavirus remains infectious 60 min after atomization (25 °C and 79% RH), while only around 5% survived 60 min at a higher temperature and lower RH (38 °C and 24% RH) (Zhang, R. et al. 2020). Also, active airborne influenza virus has been detected 90 min after aerosolization (Pyankov et al. 2018). In general, the larger the droplet size, the longer the viruses in them are expected to survive (Elsaid et al. 2021).

2.3 The Spread of Microbes Via Aerosol

Aerosols are colloidal small different sized airborne liquid, solid, or semisolid particles (Wang et al. 2021). Aerosols can be generated in countless different circumstances; plants, plumbing, showering, water-damaged materials et cetera (Prussin & Marr 2015). In addition to the aerosols created by the environment, humans generate large amounts of aerosols, especially via their respiratory tract. Since liquid evaporates into air, the liquid in aerosols evaporates over time. Droplets smaller than 50 µm in diameter usually evaporate within 1 s of emission from the relatively high humidity human body into the relatively low humidity ambient air. The liquid in aerosol droplets can contain proteins,

salts and various other organic and inorganic matter (Azimi & Stephens 2013), leading to a slower evaporation rate than water (Wang et al. 2021). They can also contain larger particles, especially biological ones (Shiraiwa et al. 2017), that can be found in abundance of varieties, sizes and amounts (Zemouri et al. 2017). These components can result in the droplet having an electrical charge or an unneutral pH value. Aerosol particles may deviate from airflow paths due to factors such as inertia, Brownian motion and external forces—including gravitational, electrophoretic and thermophoretic effects (Wang et al. 2021).

Aerosols containing biological PM are called bioaerosols and their adverse effects on health depend on the particles in them (Ghosh et al. 2015). Higher bioaerosol concentrations are associated with greater infectivity, sensitization and toxicity (Dai et al. 2021). When the fluid in the droplet evaporates, the particles left behind create a droplet nucleus, which typically has around half of its original diameter. These droplet nuclei are believed to have an average diameter of around 1–3 μm (Azimi & Stephens 2013). This partly explains why viruses are enriched in small aerosols ($<5 \mu\text{m}$) (Wang et al. 2021; Fennelly 2020). For example, infectious SARS-CoV-2 has been detected in aerosols ranging from 0.25 to $>4 \mu\text{m}$ (Wang et al. 2021). Aerosols originating from older humans usually contain a higher viral load (Fennelly 2020) and the most infectious aerosols are presumed to contain an estimated 10^9 viral RNA (Ribonucleic Acid) copies per millilitre (Bazant & Bush 2021). This also means that the particles suspended in the aerosol droplets could stay airborne as smaller droplet nuclei for much longer after most of the fluid from the initial droplet has evaporated (Nazaroff 2022). These smaller nuclei can also be influenced much more by air currents, since airborne aerosols have very similar overall aerodynamics to other PM, with size being the most important determinant of aerosol behaviour (Morawska 2006; Fennelly 2020). The number of aerosols generated by a person can vary depending on the person and their activity level. In general, the higher the decibels, the more the person creates and emits aerosols (Gregson et al. 2021; Bazant & Bush 2021). Some humans naturally produce more aerosols than others (Gregson et al. 2021). Typically, bioaerosol distribution is found to be higher in the summer (Ghosh et al. 2015).

Aerosols can be hygroscopic, meaning they tend to absorb moisture from the air, or contrarily non-hygroscopic (Joubert et al. 2010). They are tied to RH through their humidity deliquescent point. Indoor temperature and RH typically range from 21 °C to 27 °C and 20% to 70%, respectively (Zhang, R. et al. 2020). Evaporation of aerosols happens faster in higher temperatures. A higher RH is associated with increases in some aerosols falling distances and lower relative humidities result in more rapid evaporation

(Morawska 2006), leading to an increase in the abundance of suspended aerosols (Xie et al. 2007). However, mid-range relative humidities, ranging from 40% to 70%, have been observed to minimize the survival and viability of bacterial and viral organisms (Xie et al. 2007). In general, the same can be said for higher temperatures regarding viruses.

Aerosols emitted by humans are created in different ways, mainly via the respiratory tract. Respiratory aerosols are typically categorized based on their site of origin within the respiratory tract; alveolar, bronchiolar, bronchial, laryngeal and oral aerosols (Wang et al. 2021). Humans generate all sizes of aerosols, and it is believed that up to 90% of the droplets we generate are smaller than 1–2 μm (Azimi & Stephens 2013). The smaller the size, the deeper the aerosols originate in the respiratory tract (Wang et al. 2021). This is why breathing produces smaller particles than singing and speaking (Gregson et al. 2021); an average of 7200 aerosol particles per litre of exhaled air (Wang et al. 2021). For research purposes, aerosols can be generated with nebulizers and atomizers (Caddum & Donskey 2022; Morawska 2006).

Smaller droplets are believed to be more infectious (Azimi & Stephens 2013; Wang et al. 2021), since they penetrate deeper into our respiratory tract. Particles sized around 10 μm can reach the bronchial and alveolar regions, but with low deposition efficiency (Wang et al. 2021; Nicas et al. 2005; Nazaroff 2016). The Infectious Diseases Society of America defines respirable particles as having a diameter of $\leq 10 \mu\text{m}$ (PM_{10}), while larger ones are inspirable particles, that are mainly deposited in the upper airways (Tellier et al. 2019). In general, aerosols $>5 \mu\text{m}$ and $<0.001 \mu\text{m}$ deposit primarily in the nasopharyngeal region, while aerosols 0.01–0.1 μm deposit primarily in the alveolar region. The overall deposition efficiency is highest with particles $<0.01 \mu\text{m}$ and particles sized 0.1–1 μm have the worst deposition efficiency at around 0.2 (Wang et al. 2021).

For example, one normal sneeze has been evaluated to generate approximately a million droplets $<100 \mu\text{m}$ in diameter (Xie et al. 2007). A cough has a combined volume of $6 \times 10^{-8} \text{ mL}$ in aerosols that have an initial diameter $<10 \mu\text{m}$, which is around a third of the cough's total volume. A normal cough produces an average of around 75 000 particles (Prussin & Marr 2015) and a sneeze can emit around 200-fold more particles than a cough (Nicas et al. 2005). From a height of 2 m, a person's breath, cough and sneeze can carry larger droplets (60–100 μm) 1 m, 2 m and 6 m, respectively (Xie et al. 2007). In the right circumstances, aerosols can travel hundreds of meters or even more. After finally settling, for example, in laboratory testing SARS-CoV-2 has been observed to be able to survive for up to 72 h on plastic and stainless steel, 24 h on cardboard and up to 4 h on copper (Elsaid et al. 2021), meaning that they can later be resuspended into the

air by for example an air current and still cause an airborne transmission (Shiraiwa et al. 2017).

In the winter season, bacterial and fungal aerosol concentrations in different microenvironments were observed to range between 300–3650 CFU/m³ and 300–4150 CFU/m³, while in spring they ranged between 450–5150 CFU/m³ and 350–5070 CFU/m³, respectively (Kumar et al. 2022). There is a large variety of air sampling techniques with different functions (Lacey & West 2006). Active air sampling is perceived to be advantageous compared to passive air-sampling techniques, due to active air sampling providing both qualitative and quantitative data, whereas passive air sampling offers only qualitative data (Stockwell et al. 2019). However, active air sampling requires more resources, making it more expensive. In general, the most common way, due to its simplicity and affordability, consists of using some kind of trap to collect the particles, which are then studied under a microscope (Lacey & West 2006; Ghosh et al. 2015). Bacteria are often utilized in air sampling tests, since they are relatively easy to cultivate and measure in CFU. The measurement of airborne viruses in indoor environments has straggled that of bacteria and fungi, with the relatively recent developments of quantitative polymerase chain reaction and other DNA based measurement technologies facilitating and accelerating further advancements in viral load measurements (Nazaroff 2016). The progress in our understanding of air quality and its significance to humans has resulted in the development of numerous new air cleaning technologies to combat the global effects of poor air quality.

2.4 Aerobiology in Medical Facilities

The amounts and composition of PM in different medical facilities differs hugely, with endless factors contributing to them. One study found a total of around 350 substances in a medical facility's air, with 50% of them classified as of high toxicological risk (López et al. 2024). Medical facilities are known for their tendency to have higher microbial activity due to very favourable conditions for their reproduction, leading to the increased spread of pathogenic micro-organisms (Zhou et al. 2022; Nourozi et al. 2024). It is crucial to keep these facilities as sterile as possible to minimize the possibility of contaminations and infections. In general, acceptable total microbial counts (CFU/dm²/h) for medical wards, aseptic rooms and operating theatres in action and at rest can be approximated to 451–700, 51–90, 61–90 and 5–8, respectively. Lower levels can be interpreted optimal and higher levels non-acceptable (Pasquarella et al. 2000).

Infection spread in hospitals may occur by one or more of three transmission modes: contact, droplet and airborne (Stockwell et al. 2019). Healthcare workers are subject to

a considerably higher risk in acquiring pathogenic micro-organisms with a risk ratio around 2.5 for acquiring viral or bacterial infections, with many believing the ratio to be even higher (Zemouri et al. 2017). There is compelling evidence that the airborne transmission of pathogenic micro-organisms is widespread in hospital environments (Zhou et al. 2022). For example, HIV and hepatitis B have been observed to be capable to become airborne (Tellier et al. 2019). Bioaerosols can also be extremely dangerous to immunocompromised patients (Zemouri et al. 2017). This partly explains why bioaerosols are such a prevalent challenge in medical facilities (Shiraiwa et al. 2017), which in turn makes risk assessment of microbial contamination extremely important in these facilities.

However, conventional risk assessment is expensive, time consuming, resource-demanding and requires knowledge (Stålfelt et al. 2023; Mirhoseini et al. 2015). This has led to the development of novel methods of determining air contamination, such as real-time particle counters, biofluorescent particle counters and the Andersen biosampler (Stålfelt et al. 2023; Mirhoseini et al. 2016). Sampling techniques used for bioaerosols include impaction, impingement, filtration, gravity, electrostatic and thermal precipitation, cyclone and condensation. Possible enumeration techniques for culturable samples include classical microscopy, most probable number, laser-induced fluorescence, matrix-assisted laser desorption/ionization and laser-induced breakdown spectroscopy, while ones for non-culturable samples include epifluorescence microscopy, polymerase chain reaction, flow cytometry, next generation sequencing, denaturing gradient gel electrophoresis and biomarkers. Although, it must be mentioned that there are still varied results on the accuracy of different particle counting techniques (Stålfelt et al. 2023; Landrin et al. 2005; Mirhoseini et al. 2015; Mirhoseini et al. 2016; Figuerola-Tejerina et al. 2020). There are countless control mechanisms, but the combination of different methods has been found to be most effective (Ghosh et al. 2015).

Bioaerosol monitoring in indoor hospital environments has been hypothesised to be an important approach in the overall prevention of airborne healthcare-associated infections. By monitoring airborne micro-organisms in medical facilities, data on bioaerosol sources, concentration and dispersion can be extracted (Mirhoseini et al. 2015), which in turn gives information on the total microbiological activity. Air filters have been observed to be a possible way for microbial monitoring with the utilization of quantitative filter forensics (Fernández de Mera et al. 2022; Hoskin et al. 2024).

The different PM levels in medical facilities vary depending on countless different factors, such as the unique aerobiology of the surrounding location, weather, season, number of humans in the facility and its ventilation and other cleaning techniques. The quantity of different kinds of PM can also vary a lot between two different sections within the same

facility (Chamseddine et al. 2019; Wan et al. 2011; Nardini et al. 2004; Jung et al. 2015; Mohammadyan et al. 2019). For example, the I/O ratios in different areas can differ substantially; I/O ratios for PM_{2.5} can be less than 0.1 in rooms with more precaution used, while being over 5 in others (Nardini et al. 2004; Wan et al. 2011; Mohammadyan et al. 2019). Another hospital measured bacterial and fungal levels in 3 different areas (emergency service, surgery ward and operating theatre); 240–736 CFU/m³ and 27–933 CFU/m³, 99–495 CFU/m³ and 1–32 CFU/m³, 12–170 CFU/m³ and below 1 CFU/m³, respectively (Cabo Verde et al. 2015). A direct proportionality between levels of different fungi in different areas in a hospital has been observed and thus suggested (Araujo et al. 2008). Another study found the mean total highest bioaerosol concentrations in inpatient facilities (Stockwell et al. 2019).

In operating rooms, where it is crucial to keep all contaminant levels as low as possible, the levels can be for example a hundredth of the levels in different spaces where precaution is less prevalent, such as break rooms and office spaces (Nardini et al. 2004). In general, I/O ratios seem to either follow those of normal buildings (Mohammadyan et al. 2019) or they can be relatively high in medical facilities and can exceed health standards by many folds, especially in areas where less precaution is used and there are more humans in comparison to the size of the area (Chamseddine et al. 2019). Also, it has been observed that indoor CO₂ and total VOC levels in hospitals surpassed those recorded outdoors with I/O ratios >1 (Jung et al. 2015; Chamseddine et al. 2019).

In healthcare environments, there is higher potential for horizontal airflows/cross-airflows, which can relocate airborne PM, while also prolonging their airtime. In turn, downdrafts can drastically reduce airtime by pushing particles to the ground (Tellier et al. 2019), which is a concept utilized often in for example surgery wards. One study found that areas using conventional mechanical ventilation systems had around a 20 CFU/m³ bioaerosol concentration, while areas with only natural ventilation had concentrations 10-fold compared to mechanical ventilated areas. Predictably, areas using sophisticated mechanical ventilation systems, such as increased ACH, directional flow and filtration systems, had the lowest total bioaerosol concentrations of around 9 CFU/m³ (Stockwell et al. 2019).

In healthcare facilities, total VOC levels correlated well with CO₂ levels (Cadnum & Donskey 2022), which correlate with PM levels, which in turn correlate with biological particle amounts (Wan et al. 2011; Liu et al. 2018). This means that in theory we can use for example CO₂ levels to estimate the levels of some biological particles. Medical buildings are recommended to have a typically higher ACH of around 2–12 and even more depending on the gravity of the environment (Ye et al. 2022; Srikrishna 2022), since ACH

has been observed to act as a key factor in mitigating airborne pathogen transmission within healthcare facilities (Huzayyin et al. 2024; Pasquarella et al. 2000).

Gram-positive bacteria and fungi were found most often in hospital environments (Prussin & Marr 2015; Zemouri et al. 2017; Wan et al. 2011; Cabo Verde et al. 2015; Alghamdi et al. 2023). A comprehensive study evaluated that the mean $\log_{10}$ of CFU/m³ in hospital environments ranged from 0.8 to 3.8 for the bacterial load of bioaerosols, while fungal loads ranged from $\log_{10}$ 0.8 to 3.5 CFU/m³ (Zemouri et al. 2017). A similar bacterial load ($\log_{10}$ 1–3.9 CFU/m³) was noted in dental clinics. Fungal counts seem to also vary depending on the hospital, with for example one study concluding that outdoor fungal levels were considerably higher than indoors (Alghamdi et al. 2023). However, there can be a high variation in the levels between fungi and bacteria, since for example one study found that 90% of airborne micro-organisms were bacteria in different areas of the hospital (Zhai et al. 2018). In general bacteria, virus and fungi levels in medical facilities should be approached individually case-by-case, since they depend for example on specific microenvironments. However, in general the levels seem to correlate relatively well with data from normal buildings; higher bacteria load, lower fungi loads and lowest viral loads. Although, in relation virus concentrations are speculated to be much higher when compared to normal buildings.

3. HEPA FILTRATION

3.1 The Principles of HEPA Filtration

The first ever HEPA filters were designed by the U.S. Department of Energy in the 1940s as a part of the Manhattan Project to combat the accumulation of radioactive PM in the air. Although it took around a decade for them to gain widespread commercial adoption, significant advancements have since been made, leading to improved particle removal efficiency and reduced operational costs, while their fundamental design and function have remained mostly unchanged (Cheremisinoff 2016). In all simplicity, it is a filter made from fine fibres that are arranged to form a porous filter media. The fibres should be arranged so that air can flow through the media with minimal resistance, while the PM from the air is trapped in the fibres, while ideally maintaining low pressure drop and high filtration performance (Hwang et al. 2022). This results in cleaner air with a lower concentration of contaminants (Carmona et al. 2022; Kelly & Fussell 2019; BurrIDGE et al. 2024; Ebrahimifakhar et al. 2024; Yan Suen et al. 2022; Mousavi et al. 2020; Karjalainen et al. 2017; James et al. 2020; Zhou et al. 2022). Accordingly, HEPA filters are associated with lower microbial contamination loads indoors (Ting Wu et al. 2021; Araujo et al. 2008; Elsaid et al. 2021; Liu et al. 2017; Dai et al. 2021; Kompatscher et al. 2023; Conway Morris et al. 2022; Conway Morris et al. 2022; BurrIDGE et al. 2024; Mousavi et al. 2021; Liu et al. 2022; Yan Suen et al. 2022; Hoskin et al. 2024; Abdul Salam et al. 2010; James et al. 2020; Zhou et al. 2022; Curtis 2022). Only sparse studies found negligible benefits from HEPA filtration (Falkenberg et al. 2023; Ebrahimifakhar et al. 2024), with most of them reporting factors that might have hampered with the results.

HEPA filters are a subcategory of air filters, which belong to the subgroup of mechanical filtration (Liu et al. 2022). HEPA is a cohesive categorization for all air filters with a >99.95% (European standard) or >99.97% (United States of America standard) filtration efficiency against MPPS PM (Kumbargere Nagraj et al. 2020) and higher efficiencies for particles smaller and larger than the MPPS (Lelieveld et al. 2020). Both have different categorizations for air filters based on their efficiency. According to the United States of America's standards, filters are categorized according to minimum efficiency reporting values. The classifications range between classes 1–20. HEPA filters are classes 17 and 18, with Efficient Particulate Air (EPA) filters consisting of the classes 1–16 and Ultra-Low Penetration Air (ULPA) filters being classes 19 and 20 (Cheremisinoff 2016). This means that in order of increasing filtration efficiency, EPA, HEPA and ULPA filters are

categorized based on their efficiency. The same goes for the European classification, in which HEPA filters are categorized into classes H13 and H14, with EPA filters consisting of the classes up to E12 and ULPA filters being classes U15–U17. An EPA E12 filter must achieve at least 99.5% filtration efficiency for the MPPS and classes lower than that have incrementally lower efficiencies. A HEPA H13 filter must achieve at least 99.95% filtration efficiency and a H14 at least 99.995% filtration efficiency, while the best ULPA class, the U17, must achieve at least a 99.999995% filtration efficiency for the MPPS (Todea et al. 2020).

HEPA filters rely on a combination of different mechanisms. The five main mechanisms of HEPA filters for capturing PM include the sieve effect, interception, inertial impaction, Brownian diffusion and static charge effect (Cheremisinoff 2016; Elsaid et al. 2021), of which interception, impaction and diffusion are the most important. The sieve effect stops larger particles, typically $>5\text{ }\mu\text{m}$, by the filter media acting as a sieve and catching the particles since they cannot fit through the smaller pores. Interception occurs when a particle following the airflow streamlines comes within one particle radius of a filter fibre and is captured due to physical contact. Interception most commonly affects particles above $0.3\text{--}0.4\text{ }\mu\text{m}$, while being most dominant for particles sizes $0.1\text{--}1\text{ }\mu\text{m}$. Impaction usually affects the same sized particles as interception and is most dominant against particles sized $0.5\text{--}5\text{ }\mu\text{m}$. Its mechanism differs by the particle being forced into contact with a fibre due to inability to follow the trajectory of the airstream due to its size/mass, resulting in adherence. Diffusion impacts the smallest particles, $<0.1\text{ }\mu\text{m}$, with the arrangement of the fibres making the particle collide with air or other molecules, which sets the particle on a different course. This makes the particle act like a pinball inside a pinball machine, making it zigzag around between different air flows in the filter media, until it finally gets caught due to either interception or impaction. Finally, the static charge effect can capture some particles with an appropriate electrostatic charge (Cheremisinoff 2016).

The dust loading process of filtration can be divided into three stages based on pressure drop variation with dust deposition: deep filtration, transition and surface filtration. In the deep filtration stage, large particles are trapped on the filter surface while fine particles are captured within the media, resulting in smaller pores and thus increased filtration efficiency. During the transition stage, a dust cake forms on the filter surface, signalling the shift to surface filtration. In the surface filtration stage, the pressure drop increases linearly as the dust cake builds up and filtration efficiency reaches its maximum (Teng et al. 2023; Teng et al. 2022). Smaller particles have been observed to cause a smaller pressure drop during the dust loading process, while also forming a dust cake with a higher porosity (Zhang, W. et al. 2020).

Due to the different mechanics for capturing particles influencing different sized particles differently, HEPA filtration filters some particle sizes more efficiently than others. As we can see from the size ranges of the different mechanics, they do not cover particles around $0.3\ \mu\text{m}$ very effectively. This explains the narrow size range around $0.3\ \mu\text{m}$ particles that has been observed to have a lower filtration efficiency, which is the size referred to as the MPPS for HEPA filters.

The DOP (dioctyl phthalate) penetration test is a common and standard HEPA filter efficiency test that utilizes the MPPS. The method of DOP involves creating particles sized around $0.3\ \mu\text{m}$ and running the air through a HEPA filter. The concentrations before and after the filter are compared, from which the particle removing efficiency of the filter for the MPPS can be derived. DOP is an oil, which is heated to a specific temperature, where it creates the right sized particles. Even relatively small deviations in the temperature can result in different sized particles. The MPPS being very narrow, it is crucial for the particles to be close to $0.3\ \mu\text{m}$ for the true efficiency to be measured successfully. Therefore, the DOP test must be carried out with exactitude for accurate results (Cheremisinoff 2016).

Particle removal efficiency can also be assessed by three more standard methods: the weight arrestance test, atmospheric dust spot efficiency test and particle size removal efficiency test. The weight arrestance test is used for filters with lower filtering efficiencies. It involves running a predetermined weight of dust through a filter and measuring the amount of dust the filter collects, from which the efficiency can be derived. The atmospheric dust spot efficiency test is mostly used for filters with medium efficiency and consists of naturally occurring atmospheric dust being fed to the filter. The exiting airflow particle load is observed by placing a light-collecting surface near the exit, so that possible accumulations of PM can be observed on it. The particle size removal efficiency test gives efficiencies for removing particles of specific size by filters, which are incrementally loaded with synthetic dust (Cheremisinoff 2016).

The efficiency and pressure drop of the filters have been observed to strongly depend on the filter type, particle size and airflow velocity (Hwang et al. 2022). Typical pressure drop for HEPA filters is around 25–50 mmAq, which is around 245–490 Pa (Liu et al. 2017), but new improvements in technology have brought a lot of lower pressure drop possessing HEPA filters to the markets. Filters with higher filtration efficiency typically have very small pores, which trap particles that are too large to pass through. One H14 HEPA filter for example has a porosity of 99.4% and tortuosity of 99.1% (Huzayyin et al. 2024). As can be deduced, the lower efficiency filters have lower efficiencies especially with smaller particles due to larger pores. However, filters with lower filtration efficiency

can still be useful in terms of HEPA filtration when used as a type of pre-filter for the HEPA filter. Larger particles do not penetrate as deep into the HEPA filter media as they do in the case of lower efficiency filters, possibly resulting in excessive surface loading (Obitková et al. 2024). Thus, HEPA (and ULPA) filters are often protected with less expensive filters that filter out larger particles, resulting in less pressure drop (Zhang, W. et al. 2022; Todea et al. 2020). In the long run, this improves the more expensive HEPA filter's efficiency, operational costs and life-expectancy (Mousavi et al. 2021; Hwang et al. 2022).

There are many possible materials of the filtering media of HEPA filters. Typically, the filter media is a patented glass and polymer (for example polypropylene [PP], polyethylene [PE], polyvinylidene fluoride [PVDF], polytetrafluoroethylene [PTFE], and polysulfone [PS]; Sawatdee et al. 2025) fibre blend, with new filter materials, such as non-woven nanofibres, evolving constantly at a rapid rate to achieve extremely high filtering efficiencies (Liu et al. 2017; Obitková et al. 2024). PTFE filters were observed to have much lower initial pressure drop compared to glass fibre-based HEPA filter media, but their total lifetime energy consumption may increase due to the faster rate at which their pressure drop increases (Zhang, W. et al. 2022). Ultra-lightweight nylon 6–polyacrylonitrile nanofibre-nets exhibited high filtration efficiency (99.99%), surpassing results of glass and polymer fibre-based filter media (Wang et al. 2015). PVDF nanofibre-nets showed a 99.985% filtration efficiency for $PM_{0.3}$ (particles under $0.3\ \mu\text{m}$ in diameter) (Lu et al. 2021). Also, other nanofibre-based filters have shown even more impressive filtration efficiencies, such as 99.998% for $PM_{0.3}$, while still maintaining a low pressure drop (Liu et al. 2019).

Fibrous filters are typically made from fibres with diameters ranging around $1\text{--}5\ \mu\text{m}$ (Lu et al. 2021) and the filter porosity typically falls between 50–70%. This structure allows for the effective removal of both $<1\ \mu\text{m}$ and $>1\ \mu\text{m}$ particles, while maintaining relatively low airflow resistance (Podgorski et al. 2006). However, this leaves room for a phenomenon called tunnelling, where the filter media has an inhomogeneous structure, leading to the creation of favourable less resistant pathways in the filter media. Thus, air flows through the filter unevenly leading to uneven PM deposition, which can possibly result in problems in the long run (Podgorski et al. 2006; Zhang & Liu 2022). However, it has been observed that a filter with a uniform deposition has a slightly higher operating resistance compared to nonuniform deposition, meaning that a design promoting reasonable non-uniform deposition could be beneficial to prolong the filter's life-expectancy (Zhang & Liu 2022).

HEPA filters can be flat, but more often they are pleated to provide more filter material in a smaller space (Joubert et al. 2010; Cheremisinoff 2016). Pleated filters are often structurally more intricate and are either V-shaped or U-shaped (Teng et al. 2023). This is done for increased efficiency, although simultaneously making them more prone for structural damage. Sometimes the filtration media has support elements, but many filters are also support-free (Obitková et al. 2024). The HEPA filter can also be in a cylindrical shape (Zhang & Liu 2022). All filters can also have for example pinhole leaks, which can let unfiltered air through a certain hole (Obitková et al. 2024). A recommended filter design includes a triple-layer structure: a sturdy microfibre backing, a porous nanofibrous middle layer for capturing the most penetrating particles and a protective front layer to trap larger contaminants and reduce clogging (Podgorski et al. 2006), although the optimal design varies a lot on the materials used.

Electrospinning, a process that uses a high-voltage electric field to draw a polymer liquid into fine fibres, which are then used to form fine layered mats, was used to incorporate CuBTC/TiO₂/PS-based nanofibres on a commercial HEPA filter (Sheraz et al. 2023). The filter exhibited high adsorption efficiency for VOCs and with a self-cleaning UVGI (ultra-violet germicidal irradiation) strategy, it had antimicrobial effects. However, electrospinning still has many drawbacks, such as limited structural controllability and uncontrolled packing structures (Liu et al. 2019). As an alternative to electrospinning, recent advances in melt-blown technology have enabled the production of nanofibrous filter media with controlled structures and high porosity, making them highly effective for capturing sub-micrometric PM. Melt-blown filters have been observed to be more cost-effective than glass fibre filters (Hwang et al. 2022). Also, HEPA filters based on carbon nanotubes have shown promising results, with filtration efficiencies reaching >99.999% (Li et al. 2014), although this technology is still particularly expensive. Experimental results confirm that nanofibrous layers, particularly when thick and homogeneous, significantly enhance filtration efficiency and quality factor.

By using an effortless multilayer filter fabrication method, highly efficient filters can be derived from 100% natural cellulose, making them biodegradable (Sawatdee et al. 2025). These filters were reported to have an even greater particle filtration efficiency than some regular commercial HEPA filters (95.3–97.3% vs 75–82%, respectively). The structure and surface characteristics of native cellulose fibres restrict their filtration efficiency for small particles. However, employing fibrillated cellulose and nanocellulose has been demonstrated to considerably enhance filtration efficiency (Ukkola et al. 2021). Nanofoams and aerogels, which are used in HEPA filtration, have incorporated these same cellulose nanofibres, drawing growing attention due to their beneficial mechanical

features, such as low density, high porosity, chemical versatility and biodegradability (Ukkola et al. 2021).

HEPA filtration has disadvantages, such as price, pressure increase, maintenance frequency and possible pollution of the filter (Kompatscher et al. 2023). HEPA filters have a certain lifespan, which is a function of countless interconnected factors; humidity, temperature, amount and type of collected matter, usage et cetera. Microbial communities in filters are regional (Guo et al. 2020). Microbial activity coexists with PM. Dust and PM, which can be nutrient rich to microbes (Guo et al. 2020; Ribeiro et al. 2024), builds up on the filter's surface and inside the media over time. This can act as a facilitator for the formation of biofilm, mould or other microbial life under the right circumstances, which can then turn into aerosols due to air turbulence in the filter, resulting in the possible transmission of airborne microbes (Ribeiro et al. 2024). Microbes have demonstrated the ability to replicate densely in even fibreglass or polymer filters. They can produce VOCs, which are then released into the ambient air (Ribeiro et al. 2024). The common suitable growth area for mould is at $T > 18^{\circ}\text{C}$ and $\text{RH} > 92\%$ and the safe area is at $T < 0^{\circ}\text{C}$ or $\text{RH} < 70\%$ (Liu et al. 2018). Filter material can also have an impact. Cellulose fibre filters have been associated with higher fungal colonization (Simmons et al. 1997). Fungal colonization can also penetrate from one filter to another downstream with the airflow in less than 1 month, making them very important to keep under control to prevent them from spreading. This is why visual and microscopic inspection of filters could be very beneficial, which is more reliable than detecting fungal colonization and microbial activity based purely on pressure drops (Simmons et al. 1997).

Hygroscopic and non-hygroscopic aerosols can have differing effects on HEPA filters' pressure drop. RH not only has effects on microbial viability, but also the overall efficiency of HEPA filters. RH affects differently for flat and pleated filters. In general, a higher RH corresponds with a larger increase in pressure drop, which in turn results in a lower overall efficiency and higher energy consumption. Flat filters behave more predictably across different RH levels. Compared to flat filters, structurally more elaborate pleated filters are more vulnerable to deformation, especially under high RH and when filtering hygroscopic particles, which can lead to a more rapid rise in pressure drop. Non-hygroscopic aerosols have been observed to have no significant effect on the pressure drop at $\text{RH} < 80\%$. At $\text{RH} > 80\%$ and at a later clogging stage, filtration surface is reduced due to liquid interacting with the filter media, leading to a higher pressure drop. At low relative humidities, a slight reduction in pressure drop may happen with hygroscopic aerosols. But at high relative humidities ($>60\text{--}70\%$), pressure drop increases significantly due to liquid film formation, which can result in pleat collapse (Joubert et al. 2010).

3.2 The Microbiological Efficiency of HEPA Filtration

There is no question whether HEPA filtration collects microbes from the air. Even though almost all studies agree on the reducing effects of HEPA filtration on microbial loads, most studies have highly varied results on the exact HEPA filtration efficiencies regarding different microbes. Depending on the study, results for the microbiological filtration efficiency of HEPA filters vary from >99.999% to anything less than that (Cao & Zhang 2009). According to most, efficient filters can capture approximately 90–95% of bacteria in hospital environments and for viruses smaller than 1 μm , HEPA filters are recommended in healthcare facilities (Nourozi et al. 2024). For example, SARS-CoV-2 and other viruses have been identified on HEPA filters (Ting Wu et al. 2021; Obitková et al. 2024; Clarke et al. 2023; Fernández de Mera et al. 2022), proving that they decrease virus concentration in the air. Pathogens undergo natural decay. If they are trapped inside a filter media, they decay there and are unable to cause airborne transmission (Zhang et al. 2024). However, there are varied opinions on whether HEPA filtering alone is sufficient in keeping all microbial loads low enough to prevent transmission. It is argued that lowest chances for transmission are achieved with high-quality masks combined with regular ventilation (Lelieveld et al. 2020), although having high-quality masks for everyone continuously is rather infeasible.

To achieve enhanced efficiency, reduced energy consumption and minimized by-product pollution, the advanced air filtration technologies incorporate multiple methods in a synergistic manner (Liu et al. 2017). However, they usually have at least one critical drawback; poor mechanical strength, large pressure drop, lower filtration efficiency et cetera. They can also often be very costly, which is usually their biggest overall drawback (Ribeiro et al. 2024). Correspondingly, HEPA filtration can be combined with other antimicrobial technologies to achieve a higher microbiological efficiency (Azimi & Stephens 2013). Certain materials exhibit superior antimicrobial activity compared to others, attributed to factors such as chemical stability, catalytic properties and improved thermal conductivity (Huzayyin et al. 2024). For successful incorporation into the filter media, the material should not hinder filtration efficiency, damage the filter structure or cause a large increase in pressure drop. Different substances can have different ways for inactivating microbes and do so at varied levels of success depending on the microbe, meaning one material can have drastically different effects on two different microbes (Ribeiro et al. 2024). They can either be contact-killing or release-killing, depending on if they themselves inactivate microbes or if they release another substance, that does the inactivation. Instead of direct inactivation, they can also hinder the microbial replication process, eventually leading to lower microbial levels. Analogously, incorporating a combination of

many elements with different antimicrobial properties can enhance the overall antimicrobial activity. However, the different substances and elements used can often be harmful to humans, in which case it is vital they do not detach from the filter and release into the ambient air.

While being a relatively low-cost solution for enhanced antimicrobial effects, the coating of HEPA filters, depending on the filter type, coating technique and substance used, can influence filtration efficiency and pressure drop (Ueki et al. 2022). Typically, the most used antimicrobial coatings seem to have minimal negative effects on these factors, while some have even reported positive effects from coating (Ueki et al. 2022; Yan Suen et al. 2022; Ribeiro et al. 2024). For example, silver, copper and zinc can be applied to HEPA filters by either integrating it as part of the fibres or applying a post-manufacture oxide coating on them. Since they have antimicrobial properties, they can inactivate microbes in the filter at a much faster rate compared to natural destruction (Ribeiro et al. 2024; Liu et al. 2017). CuO and Ag₂O presented a 99% viral reduction in 24h. CuO spray coating exhibited the highest toxicity to cells (such as bacteria), followed by the ZnO spray coating, with the Ag₂O spray coating showing the least toxicity. Consequently, silver, copper and zinc have been shown to cause toxic effects to human cells, but the study also indicated preliminary reliability and safety of the coating (Ribeiro et al. 2024). A HEPA filter coated with a monovalent copper compound demonstrated higher viral capture efficiency than an uncoated HEPA filter, achieving capture rates of 90.35%, 98.34% and 99.99% at filtration times of 5, 10 and 35.5 minutes, respectively (Ueki et al. 2022). In one study, silver nanoparticles were incorporated into electrospun SiO₂-TiO₂ nanofibres to create a filter with a high PM_{2.5} filtration efficiency and antibacterial capabilities (98.84% and 95.8%, respectively) (Wang et al. 2019). An antimicrobial HEPA-CIO₂ (HEPA filter coated with micro-encapsulated chlorine dioxide) has shown bacterial and viral inactivation of over 99% within 1 minute, while for spores the same 99% reduction was achieved within 30 min (Yan Suen et al. 2022). The HEPA-CIO₂ was perceived to be safe according to in-vitro and in-vivo studies, which can normally be a problem with coating.

Viruses are typically much smaller than the 0.3 µm MPPS, meaning they can pass through the filter media with a higher chance compared to particles larger than the MPPS (Obitková et al. 2024). This has resulted in filters with improved virus capture capabilities through affinity binding (Kim et al. 2021). Various antiviral and virus-adsorbing materials, such as sialyllactose, curdlan sulfate, poly(ethylenimine) and sodium chloride, have been integrated into or coated onto filters. However, these materials face limitations, including

high costs, narrow-spectrum efficacy, multi-step production processes, potential toxicity and inadequate stability (Kim et al. 2021).

Tannic acid, a polyphenol derived from plants, is a promising candidate for filter functionalization due to its antiviral properties and its potential as a cost-effective adhesive for various materials. Unmodified HEPA filters expressed a <10% virus capture efficiency, while tannic acid coated ones had efficiencies >80%, with the viruses remaining firmly attached to the coated filter surface (Kim et al. 2021). However, other studies report higher viral capture efficiencies for unmodified HEPA filters of 30–50% (Burridge et al. 2024) and 99.997–99.999% (Zacharias et al. 2021). Viral capture ratios have also been observed to correlate with the ACH, with higher ACH rates leading to greater viral load reduction (Ueki et al. 2022). Other promising organic coating solutions include for example *Eriobotrya Japonica* nanoparticles, tea tree oil and *Sophora Flavescens* nanoparticles, which all showed varied high antimicrobial reductions of up to 99%, depending mainly on contact times (Hwang et al. 2015).

Photocatalytic oxidation (PCO), a form of photocatalysis that generates oxidants, is increasingly utilized in the HVAC industry to inactivate airborne micro-organisms (Kompatscher et al. 2023). Photocatalyst materials, such as TiO_2 , possess antimicrobial traits when activated by light (typically ultraviolet) (Liu et al. 2018; Chuaybamroong et al. 2010; Luna et al. 2008). A study examined and compared Ag, TiO_2 and Ag/ TiO_2 nanomaterials deposited on a HEPA filter. The results showed that, compared to unmodified filters, the filters have 100% biocidal action in passing air and 55.6%, 72.2%, and 81% of inhibition to microbial growth in the filter with Ag, TiO_2 and Ag/ TiO_2 , respectively (de Freitas Rosa Remiro et al. 2021). This correlates with other studies that have observed PCO inactivating 60–100% of microbes depending on the microbe (Chuaybamroong et al. 2010). This explains PCO's popularity, even though factors like high humidity can have a negative impact on the results. Antimicrobial photosensitizer dyes, which are also activated by light, rely on a process similar like that of photocatalysts. For example, crystal violet-embedded polyacrylonitrile nanofibres electrospun onto HEPA filters exhibited >99.9% antibacterial efficiency within 2 hours under white light exposure (Lee et al. 2025).

UVGI can be utilized with HEPA filtration by installing it to radiate ultraviolet light on the filter. This combination has been observed to be very effective (Huzayyin et al. 2024; Kompatscher et al. 2023). The destructive inactivating effect of UVGI on bacterial and viral DNA and RNA depends on the microbe and intensity of radiation. There are three subcategories of UVGI based on irradiation wavelength: UV-A (320–400 nm), UV-B (280–320 nm) and UV-C (200–280 nm). Out of these, UV-C is thought to be the most efficient at inactivating viruses, bacteria and spores (Huzayyin et al. 2024; Luna et al.

2008). Since viruses do not have cell walls, they are more easily penetrated by ultraviolet light and inactivated than fungi and bacteria (Kompatscher et al. 2023). Although, for example Chinese guidelines do not recommend the use of UVGI, because of its potential radiation hazards (Ye et al. 2022). One study revealed a reduction of 50% with UVGI compared to 65% with cold plasma. Including a HEPA filter increased these reductions to >99% and >95%, respectively (Kompatscher et al. 2023).

Cold plasma is a partially ionized gas with electrons at high energy, but the gas remains near room temperature. They can inactivate micro-organisms by for example cell wall destruction, DNA/RNA damage, protein denaturation and oxidative stress. Ozone, which can be produced by cold plasma, interacts with the bacterial cell membrane, potentially through the ozonolysis of carbon–carbon double bonds in membrane lipids, leading to cell lysis. Laboratory studies have demonstrated that ozone concentrations exceeding 100 ppm, particularly in high-humidity environments, exhibit strong virucidal activity against RNA viruses. Due to their high reactivity, ozone molecules readily interact with micro-organisms, neutralizing them. However, concerns have been raised regarding the ozone concentration required to effectively eradicate pathogens in the air and the potential health risks to dental personnel and patients. Nevertheless, ozone has a half-life of approximately 20 minutes and decomposes into oxygen, mitigating its potential health hazards (Kumbargere Nagraj et al. 2020).

Microbes can carry positive or negative charges. Thus, they can be captured with an opposite polarity charge. Electrostatic, or Columbic filtration captures particles by using an electrostatic field that attracts charged particles to oppositely charged surfaces. In contrast, dielectrophoretic filtration captures both charged and neutral particles by applying a non-uniform electric field that induces a polarization force on particles, causing them to move toward areas of higher field intensity (Wang & Otani 2013). The antimicrobial activity of charged filter fibres has been studied, and it was concluded that electrical interference reduced filtration efficiency and increased pressure drop (Leung & Sun 2020). Ionizers employ charged electrodes to release negative ions into the air. These ions transfer electrical charges of the same polarity to aerosol particles, inducing repulsion between particles and causing them to move away from the breathing zone, ultimately settling on surrounding surfaces. Also, an alternative proposed mechanism indicates that micro-organisms suspended in the air are attracted to these negatively charged ions, becoming heavier and thereby precipitating onto surfaces (Kumbargere Nagraj et al. 2020). However, while air ions can damage microbial cell membranes through physical or chemical interactions (Kim et al. 2011), this process does not fully

destroy the micro-organisms. As a result, further treatment using conventional disinfection methods may be necessary to ensure complete inactivation (Kumbargere Nagraj et al. 2020).

Currently, formaldehyde fumigation is rarely employed due to the carcinogenic risks associated with formaldehyde, with fogging emerging as a preferred alternative. Fogging involves the use of a mixture of hydrogen peroxide and silver ion solution, which effectively controls contaminated aerosols through its bactericidal action. Air purifiers can also utilize an activated carbon filter and a HEPA filter together. The carbon filter is ideal for removing chemicals and odours in the air, while HEPA is ideal for the particles (Kumbargere Nagraj et al. 2020). Adsorbents, such as activated carbon and hydrophobic zeolites, eliminate gaseous pollutants (for example VOCs, O₃, NO₂, SO₂ and hydrogen sulphide) (Kumar et al. 2023). However, they typically need a lower RH to work properly.

3.3 HEPA Filtration in Medical Facilities

In the modern era, almost all hospitals have some type of air-conditioning system (Ting Wu et al. 2021). In hospitals, a central air condition system can on average remove more aerosol pollutants than a non-central air condition system (Jung et al. 2015). They are recommended to supply 100% outdoor air, meaning air recirculation should be avoided (Mousavi et al. 2021) and they should be operated for around 2 h before and after occupation in areas with activity only during the day (Ye et al. 2022). Filtration is a key method employed in healthcare facilities to eliminate airborne pathogens from the air (Emmerich et al. 2013). One study evaluated that the protection provided by air cleaners was equivalent to more than 30 ACH for the healthcare workers (Nourozi et al. 2024). HEPA filters can either be utilized as a part of the ventilation system or in a separate air cleaning unit. HEPA filtration is highly recommended in medical facilities (Ye et al. 2022), with isolated studies even concluding that they are gratuitously efficient in some cases, making lower efficiency filters a more viable option (Mousavi et al. 2021; Nishimura et al. 2022; Whyte et al 2012). In areas where it is paramount to keep microbial loads low, such as intensive care units and operating rooms, HEPA filters are often integrated as part of the ventilation system (Ting Wu et al. 2021; Mousavi et al. 2021). Air conditioning system contamination has been linked to the transmission of certain hospital-acquired infections (Ting Wu et al. 2021).

Ventilation systems' air supply and exhaust components are recommended to be equipped with lower and higher efficiency filters, especially for negative pressure areas, such as isolation (Ye et al. 2022; Mousavi et al. 2021). Ventilation systems for many hospital areas are often designed to circulate air with a laminar flow, which means that

air flows in a certain direction, usually pushing contaminated air towards the ground. HEPA filtering combined with laminar airflow has proven great results and thus is recommended in the most critical areas (Mousavi et al. 2021). Although, there is evidence that temperature-controlled airflow and turbulent airflow might remove aerosols better than laminar airflow in some cases (Dai et al. 2021; Izadyar & Miller 2022). A positive airflow, which is used in for example positive pressure isolation wards, combined with HEPA filtering has also been observed to have a positive effect (Araujo et al. 2008).

Depending on the medical facility in question, acceptable microbial loads vary greatly. This means that some facilities do not necessarily need HEPA filtration to keep microbial loads at an acceptable level. Since HEPA filtration is relatively costly, having it installed everywhere in a large medical facility will be a large expense. However, the energy cost of running HEPA filters is likely to only be around 1–2% of a room's heating costs (Burrige et al. 2024). It can be argued that HEPA filtration is an expensive solution, but there are also studies, that concluded the monetized health benefits of filter installations outweighed the operation costs by up to a factor of 10 (Kelly & Fussell 2019). Although they are expensive, they can be still the cheapest option in some cases. For example, rooms with HEPA-filtered systems were observed to be a better cost-saving intervention than antifungal prophylaxis and environmental protection measures in prevention of invasive aspergillosis (\$2 665, \$42 531 and \$4 073, respectively) (Ting Wu et al. 2021). A separate meta-analysis indicated that the implementation of HEPA filters in hospital rooms was associated with a 71% reduction in invasive fungal infections (Curtis 2022).

Some of the most popular substitutes for HEPA filters in parts of hospitals where there is no filtration are lamps with germicidal ultraviolet irradiation, increasing room ventilation rates and a less widely applied generation of hydrogen peroxide mists stabilised with silver ions (Ting Wu et al. 2021). However, the fitting of HEPA filters into ventilation systems is not the only way for having HEPA filtration. Although ceiling-mounted HEPA filtration units perform better than portable units (Emmerich et al. 2013), portable HEPA air cleaners have also presented extremely positive results in decreasing microbial loads and other contaminants in the air in cases, for example, where the ventilation system cannot secure adequate air exchange (Ting Wu et al. 2021; Kumbargere Nagraj et al. 2020; Emmerich et al. 2013; Nourozi et al. 2024; Thuresson et al. 2022). For example, Methicillin-Resistant *Staphylococcus aureus*, more commonly known as MRSA, was reduced 75–93% with portable HEPA units using relatively lower ACH rates and suboptimal positioning (Boswell & Fox 2006). Although they are not designed for large areas, multiple units can be used at once to achieve a higher efficiency in a larger space. They

also have many positives when compared to ventilation systems, such as easier maintenance, manoeuvrability and lower adaptation costs. The cost of widespread portable HEPA filtration in hospitals has been predicted to save money in the long run due to decreases in nosocomial infections (Abdul Salam et al. 2010). However, with portable air cleaners there are factors to consider in their operability that can directly influence the filtration efficiency, such as positioning and efficaciousness with surrounding airflows (Saw et al. 2022; Qian et al. 2010).

According to one study, the two following conditions are necessary for the successful use of a portable HEPA cleaner in a hospital ward (Qian et al. 2010). Firstly, “the portable HEPA air cleaners should recirculate all or nearly all room air through the HEPA filter”. Secondly, “in cases where the airflow patterns may be important (such as isolation rooms), the portable HEPA air cleaners should not disturb the existing airflow pattern produced by the existing ventilation system”. In general, higher ACH is typically better. Although, the highest ACH does not always guarantee the best results with portable HEPA units, since the airflows created by them can interact with existing airflows in different ways (Bluyssen et al. 2021). So, portable HEPA units must be applied case-by-case, since correct and optimal implementation depends on the environment (Dai & Zhao 2022; Bluyssen et al. 2021; Tsang et al. 2023). In one case with a rectangular room, infection probability of less than 10% required locating the portable air cleaner in the centre of the room (and a sufficient ACH) (Dai & Zhao 2022), while another study found that the best results (99.46% microbial decontamination efficiency) were achieved when the portable unit was in the middle of one of the lateral sides of a chamber (Moradi Kashkooli et al. 2020). One study concluded that the total efficiency of the portable HEPA unit for medium exhalation was 16% to 36.44% and 48.81% from a single unit, multiple units placed on the floor and multiple units placed 1 meter above the floor level, respectively (Qian et al. 2010). Results for weak exhalation corresponded with these results. Incorrect implementation of portable HEPA units can have drastically negative effects. For example, a portable air cleaner disrupting the surrounding airflows can redirect contaminated air towards patients and personnel (Mousavi et al. 2021).

Airflows and pollutant distributions can be studied, for example, experimentally with non-harmful smoke or theoretically with computational fluid dynamics (Mousavi et al. 2021; Dai & Zhao 2022; Moradi Kashkooli et al. 2020). These can also be used to assess the most optimal place for a portable HEPA unit. An effective air change rate for portable cleaners can be calculated theoretically with the use of simulation models by using the airflows and layout of the room as parameters (Qian et al. 2010), but it can also be experimentally tested by observing airflows, or by evaluating microbial activity. A type of

patient isolation room can be created with a portable HEPA unit, by placing the unit in an anteroom at the entrance, where it can clean the air inside the patient room, while creating negative pressure with respect to the anteroom (Mousavi et al. 2021). Inside an isolation room, the most effective place was next to the patient's bed (Mousavi et al. 2020). Portable HEPA units can also, for example, incorporate antimicrobial ZeBox technology, which has been observed to maximally reduce the mean airborne microbe count by 95% and mean surface microbe count by 90% (Nagaraj et al. 2022; Narayan et al. 2022). ZeBox has several advantages compared to other microbe inactivation technologies; effective against a wide range of different microbes with no harmful chemical emissions and low energy utilization (Narayan et al. 2022).

One study reported observations of a portable HEPAiRx unit reducing PM_{10} , allergen, endotoxin and dust mite concentrations on average by 72%, 78%, 44% and 71%, respectively. In addition, the unit was observed having a possible correlation with decreases in concentrations of organic and inorganic gaseous pollutants. A reduction of 59%, 19% and 30% was observed in VOC, CO_2 and CO concentrations, respectively. However, HEPA filters alone do not have the possibility to filter out gaseous pollutants, meaning the reductions were caused by additional ACHs diluting the pollutants in the air. Also, exhaled breath condensate nitrate concentrations were lower when the HEPAiRx was in operation. HEPA filtration was also associated with a prominent improvement in sleep quality. These results indicate that HEPA filtration can be useful in reducing symptoms in asthma patients, which is supported by other studies. Portable HEPA units, like the HEPAiRx, can also be equipped with an energy recovery ventilator, which recovers energy, resulting in reduced operational costs (Xu et al. 2010).

Due to used HEPA filters possibly containing dangerous pathogens, they should be changed and dealt with accordingly. They should be collected and disposed of as medical waste or undergo thorough disinfection to prevent the risk of secondary contamination (Rodríguez et al. 2021). Physical treatments, such as ultrasonic, washing, microwave, heat and electrophoresis have also been tested to clean HEPA filters (Cheng et al. 2021). The results varied, with washing giving the best results. However, physical treatments can have negative impacts on the filter. Thus, a filter that is cleaned will most likely not have an as efficient filtration as a new one would. Since most filters are non-biodegradable, they should not end up in the nature. However, studies have reported that approximately 10 million air filters are improperly disposed of and released into the environment each month (Sawatdee et al. 2025).

4. STANDARDS AND REGULATIONS OF HEPA FILTRATION IN MEDICAL FACILITIES

4.1 Significance

The first official ventilation standard took effect over 50 years ago (Persily 2015). However, there were already guidelines well before this, which were and still are less formal compared to standards. Guidelines are usually not legally binding, while standards can be legally binding, depending on the organization compiling them. Guidelines come from for example WHO, European Centre for Disease Prevention and Control (ECDC) and government run agencies. Standards come from for example the International Organization for Standardization (ISO), European Union (EU), European Committee for Standardization (CEN) and American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), which, contrary to its name, is an international society. Legally not binding standards and guidelines can often still be perceived as mandatory in a sense that the organization needs to meet insurance, accreditation or safety criteria, which are often in line with guidelines and standards. Regulations come from governments or authorized agencies and are legally binding. Thus, they must always be followed.

Since the first standards and guidelines, the IAQ and ventilation standards have developed gradually but surely (Persily 2015). However, the continuous development of humankind usually outruns the pace at which new standards are implemented. This calls for the need to keep the standards as up to date as possible, meaning it is a continuous flexible process and never ready. Medical facilities must continuously adapt to new guidelines, standards and regulations.

4.2 Finland

In Finland, primary guidelines, standards and regulations concerning filtration in medical facilities come from the following entities: CEN, ISO, EU, WHO, ECDC, ASHRAE, the Finnish government and the Finnish Institute for Health and Welfare (THL). The Finnish Standards Association (SFS), National Supervisory Authority for Welfare and Health (Valvira), Qualification Requirements in the Building Sector (FISE) and Finnish Safety and Chemicals Agency (Tukes) are all entities that are additionally directly or indirectly linked to these requirements.

Published by CEN, the European norm EN 1822 standard specifies the classification, performance and testing of EPA, HEPA and ULPA filters (Camfil 2021). The international

standard ISO 29463 was derived from EN 1822, and they share many of the same classifications, including the ones for H13 and H14 (Camfil 2021). For example, ISO 29461 and ISO 29462 also touch upon the use of HEPA filters. In turn, the EU adopts standards like EN 1822 and ISO 29463 for classifying and testing HEPA filters.

WHO emphasizes the importance of adequate ventilation in healthcare settings to reduce the risk of airborne infections (WHO 2021). They use both EN and ISO standards and recommend the incorporation of HEPA filters into HVAC systems and the use of portable HEPA cleaners. The ECDC has very similar recommendations to WHO (ECDC 2020). They recommend the use of HEPA filters in high-risk areas within healthcare facilities, especially during respiratory virus outbreaks. They also suggest a RH of 40–60% to limit the survival of viruses. The ASHRAE Standard 170 specifies ventilation requirements for healthcare facilities, including the use of HEPA and other comparable filters in specific areas (ASHRAE 2021).

The Finnish government supports efforts to enhance IAQ in public buildings, including hospitals. The government entity Ministry of Social Affairs and Health of Finland, which prepares laws such as the Health Protection Act and the regulation of hospital facilities, oversees THL and Tukes. They for example have a program aiming to improve IAQ in various environments, including healthcare facilities through THL (THL 2021). The SFS, which serves as Finland's national standards entity, adopts international standards like ISO 29463 and EN 1822 for air filtration (SFS 2025). These standards are applicable in Finnish healthcare settings. Valvira oversees health protection laws in Finland (Valvira n.d.). Operators must notify municipal health protection authorities when establishing or commissioning facilities that may impact health, which includes considerations for air quality and ventilation systems. FISE certifies professionals in the building services sector in Finland, ensuring that professionals are qualified to design and maintain systems that may include HEPA filtration (FISE n.d.). Tukes supervises the safety of technical devices and chemicals in Finland. While specific guidelines on HEPA filters in medical facilities are not detailed, Tukes ensures that equipment used complies with safety standards (Tukes n.d.).

In general, there is not much legally binding guidance for the use of HEPA filters in medical facilities in Finland. Most just emphasize on clean indoor air in general and give a standard threshold. No uniform international standards for acceptable limits of bioaerosol loads have been established (Ghosh et al. 2015). Also, no international consensus exists on bioaerosol sampling methods and the frequency of sampling in hospitals (Wan et al. 2011).

4.3 Germany

To have a comparison for the guidance in Finland, Germany, a country known for strict regulations, was chosen. The entities CEN, ISO, EU, WHO, ECDC and ASHRAE have a similar influence on the guidance on HEPA filtration in medical facilities in Germany as in Finland. They build the so-called framework, on which the regional regulations are drafted. The rest of the entities with an influence on the guidance on HEPA filtration in medical facilities in Germany include; German Institute for Standardization (DIN), Association of German Engineers (VDI), Germany's national public health institute Robert Koch Institute (RKI), Federal Institute for Drugs and Medical Devices (BfArM), Federal Institute for Occupational Safety and Health (BAuA), Germany's central public health law (IfSG), German Society for Hospital Hygiene (DGKH), German Social Accident Insurance (DGUV) and Authority of the States for Safety Technology (ZLS).

DIN has the same role in Germany as SFS has in Finland (DIN 2025). The VDI issues guidelines such as VDI 6022, which set hygiene requirements for ventilation and air-conditioning systems, indirectly influencing the use and maintenance of HEPA filters in healthcare environments (VDI 2018). The RKI provides binding hygiene recommendations for hospitals, including guidance on air quality and the use of HEPA filters in areas like operating rooms and isolation units (RKI 2024). The BfArM regulates medical devices, including air purification systems that may incorporate HEPA filters, ensuring their safety and efficacy in medical applications (BfArM n.d.). The BAuA conducts research and provides guidance on workplace safety, including air quality standards that impact the implementation of HEPA filtration in healthcare settings to protect staff (BAuA n.d.). IfSG mandates infection control measures in healthcare settings, including adherence to hygiene standards that often necessitate the use of HEPA filters in specific areas (IfSG n.d.). The DGKH issues expert recommendations and position papers on airborne infection control, ventilation and HEPA filtration, which, while not legally binding, are widely respected and used in compliance cases (DGKH n.d.). The DGUV publishes occupational health and safety regulations relevant for healthcare workers, including air quality and exposure to airborne pathogens, influencing the requirement for HEPA filtration in employee protection settings (DGUV n.d.). The ZLS coordinates market surveillance and product safety between German states, ensuring air purification devices, including those with HEPA filters, comply with product safety standards (ZLS n.d.).

In conclusion, like in Finland, there are few regulations surrounding the use of HEPA filtration in medical facilities in Germany. However, Germany's guidelines and standards make HEPA filtration de facto mandatory in key areas, meaning it is not explicitly required by law, but is effectively obligatory in practice because of strong indirect pressure from

the different entities. This same de facto mandate could be observed in Finland, but in a much smaller and lighter scale. In a nutshell, HEPA filtration in medical facilities in Germany could be perceived to be mandatory in key areas, while in Finland it is more like a recommendation.

5. CONCLUSIONS

5.1 Summary

This thesis examined the microbiological efficiency of HEPA filtration and its role in medical facilities as a critical component of airborne infection control. A broad, systems-level approach was taken by analysing not only HEPA filters themselves, but also the underlying factors contributing to microbial contamination in healthcare environments. By studying these interconnected elements, such as particle dynamics, ventilation design, environmental conditions and microbial behaviour, a comprehensive understanding of HEPA filtration in practice was achieved. This framework enabled a well-informed evaluation of HEPA filters' general overall ability to remove airborne micro-organisms, including bacteria, viruses and fungal spores, based on credible studies, while also clarifying how their performance can be either hindered or enhanced by various contributing factors. Effective HEPA implementation depends not only on filter efficiency but also on the cumulative influence of broader factors, including for example auxiliary technologies, ventilation system, maintenance routines, risk assessments and environmental monitoring.

HEPA filters, as defined by EN 1822, achieve removal efficiencies of $\geq 99.95\%$ for airborne PM of the MPPS ($\sim 0.3 \mu\text{m}$), while removing nearly all larger and smaller particles. This results from the various capture mechanisms, which are each more effective at specific particle size ranges. However, the overlap of capture mechanisms is minimized at the MPPS, making it the most challenging PM size to filter. Numerous studies confirm the filters' effectiveness in capturing PM, including bioaerosols and microbes, which again include pathogens such as SARS-CoV-2, which are significantly problematic in medical facilities.

In the case of all three microbe groups (spores, bacteria and viruses) the studies had varied results from very low microbiological efficiencies to very high ones. In the case of fungi and bacteria, the clear majority of studies agreed on the efficiency being high and only a fraction of studies concluded on an efficiency under 50%. According to this observation, we can give an estimate of the microbiological efficiency of unmodified HEPA filtration for fungi ($\sim 70\text{--}100\%$) and bacteria ($\sim 60\text{--}100\%$). However, in the case of viruses, the studies seemed to be very conflicting. Some studies reported negligible or no viral capture efficiency at all, while others reported an efficiency very close to 100%. Just

based on the capture mechanics, it can be assumed that HEPA filtration at least decreases viral loads with some efficiency, but due to the conflicting studies it is not justifiable to give an educated estimate of the efficiency.

However, microbial viability is not directly eliminated by filtration alone, but rather through the microbes becoming trapped in the filter and gradually inactivated over time. This makes supplementary technologies like UVGI, PCO and antimicrobial filter coatings valuable additions in high-risk areas to expedite microbial inactivation. In addition, continuous technological advancements like the development of new filter media aim to achieve higher filtration efficiencies, while minimizing drawbacks like cost, pressure drop and limited microbial inactivation.

The research further emphasized the importance of proper system integration. HEPA filters should be installed at least as terminal filters in critical priority areas and supported with pre-filtration to extend filter life and maintain efficiency. In facilities lacking HVAC-integrated HEPA filtration, portable air purifiers offer a viable alternative. However, their successful operating requires that they are positioned correctly, operated continuously and maintained according to protocols.

Environmental conditions were identified key factors influencing both microbial survival and filter performance. RH should be maintained between 40–60% to reduce pathogen viability while preventing mould growth and degradation of filter media. Indoor temperatures between 21–24 °C provide a balance between occupant comfort and limiting microbial proliferation, especially when combined with sufficient air exchange rates tailored to room function and infection risk.

The thesis also illustrated the guidelines, standards and regulations related to HEPA filtration in medical facilities in Europe, with an emphasis on Finland and, for contrast, Germany. CEN, ISO, EU, WHO, ECDC and ASHRAE form the foundation for both Finland's and Germany's internal framework. In Finland, the internal framework has rendered the use of HEPA filters in medical facilities a recommendation, whereas in Germany, the corresponding structure makes their use de facto mandatory in high-risk areas. However, explicit legal obligations remain limited in both countries.

In conclusion, HEPA filtration remains one of the most effective methods for reducing airborne microbial contamination in medical environments. Its microbiological efficiency can be maximized through thoughtful system design, adherence to environmental parameters and the incorporation of complementary technologies. As airborne pathogens continue to pose challenges in healthcare, the findings of this thesis support a multi-layered approach to indoor air safety that prioritizes both patient and staff health.

5.2 Recommendations for HEPA Filtration in Medical Facilities

5.2.1 Regulatory Compliance

All HEPA filters used in medical facilities should conform to applicable international and national standards. In Europe, filtration should adhere to EN 1822 to ensure $\geq 99.95\%$ efficiency at the MPPS ($\sim 0.3 \mu\text{m}$). In Finland, guidelines, standards and regulations from THL, SFS, Valvira, FISE and Tukes should be followed. In critical zones, filters should be classified at minimum as H13 and validated with tests, such as DOP, before commissioning.

5.2.2 Integration with Ventilation

HEPA filters should be installed at least as terminal filters in ventilation systems to avoid contamination from ducts. Ideally, both ends of a duct would have a HEPA filter. Regular HVAC maintenance should be ensured. For maximum microbiological control, ventilation systems should deliver $\geq 12\text{--}20$ ACH depending on room function and infection risk. Unidirectional flow, such as laminar or turbulent flow systems should be used in spaces requiring sterile environments to prevent recirculation or cross-contamination. In facilities using ventilation without HEPA filters, it should be supplemented with portable HEPA units to meet air purity standards and minimize infections.

5.2.3 Priority Areas

In general, immunocompromised patients must be prioritized, as they are the most vulnerable to airborne infections. Priority target areas for HEPA implementation include operating rooms, surgical suites, intensive care units, cancer wards, isolation rooms, dental clinics, intensive care units and other busy areas with high aerosol levels, such as staff lounges and break rooms. If possible, widespread use of HEPA filters for all medical facilities should be implemented, which can also produce savings in the long run by minimizing costs caused by infections.

5.2.4 Microbiological Filtration Efficiency and Enhancements

HEPA filters effectively reduce airborne bacteria, viruses (including SARS-CoV-2) and fungal spores, with documented efficiencies up to $>99.999\%$. Filters should be selected

or enhanced based on microbe-specific characteristics and contact time requirements. Microbe capture and inactivation should be enhanced by combining HEPA with other technologies, such as antimicrobial coatings (for example Cu, Ag and ClO_2), UVGI (UV-C 200–280 nm) systems and/or PCO (TiO_2), while ensuring ultraviolet and chemical exposures are safely shielded from occupants.

5.2.5 Portable HEPA Purifiers

Portable HEPA units should be deployed in areas without integrated filtration, particularly in older buildings or temporary wards. Their placement should be determined on a case-by-case basis, considering the surrounding environment and any existing ventilation systems to ensure effective airflow and contaminant removal. Usually centrally located placements are the most efficient. Units should not obstruct existing airflows, and they should avoid creating short-circuiting airflow, where clean air re-enters the intake. The airflow should be tested, when possible, either by computational modelling or real-life experimentation. They should be kept in operation continuously in high-risk areas; just like a ventilation system integrated filter would also continuously clean the air. They should be used according to the manufacturer's instruction manual and cleaned periodically with a similar filter change period to that of the ones integrated into ventilation systems.

5.2.6 Maintenance Protocols

Routine inspections of filters should include microscopic or molecular analysis, especially in pathogen-sensitive areas. Pre-filters should always be used upstream of HEPA filters to extend service life and reduce biofouling. The effective combining of technologies with HEPA filtration can prolong the filter's lifespan, but exact increases are hard to define without testing. Filters without auxiliary components should be changed at the very least every 12 months, or earlier based on pressure drop increases, routine inspections (for example once every month), signs of microbial growth (especially when $\text{RH} > 70\%$) or separate manufacturer's instructions. Used filters should be disposed of as medical waste.

5.2.7 Indoor Environmental Conditions

The most important environmental factor, which is RH, should be maintained between 40–60%, as this range effectively reduces the viability of both viruses and bacteria, supports optimal HEPA filter performance and maintains acceptable comfort and health conditions for occupants. Indoor temperatures should be maintained between 21–24 °C, which is a range that balances occupant comfort with infection control. Maintaining temperatures within this range combined with optimal RH helps avoid mould growth.

5.2.8 Seasonal Adjustments

Humidification should be used to avoid RH falling below 30% (for example in the dryer winter season), which could affect filter electrostatic function, hinder occupant conditions and prolong aerosol suspension, since the aerosols become smaller due to the dryer surrounding air. Warmer and more humid seasons increase microbial aerosol loads. To combat this, the filter monitoring frequency and ACH should be increased.

5.2.9 Energy Efficiency

Pressure drop should be kept as low as possible. Ideally, pleated filters should be used to maximize surface area with minimal resistance. HEPA filtration should be combined with automated fan controls or air quality sensors to balance filtration efficiency and energy demand. To prevent overloading of HEPA filters with large particles, pre-filtration stages should be used.

5.3 Discussion

This thesis set out to examine the microbiological efficiency of HEPA filtration in medical environments by exploring not only the filter technology itself but also the broader factors that influence its performance. The findings support existing evidence that HEPA filters provide highly effective removal of airborne microbes. However, as emphasized, particle capture does not equate to microbial inactivation. This is an important distinction since viable airborne pathogens can still pose transmission risks post-capture. The analysis underscored that HEPA filtration alone may not be sufficient in high-risk healthcare environments unless paired with supporting technologies, which aligns with other research

suggesting that hybrid air purification systems offer superior protection. More strict regulations could and should be set for the use of HEPA filters in medical facilities in Finland.

Even though credible sources were used, one of the major challenges encountered during this research was the inconsistency in published data. While HEPA filter efficiency is well-documented in terms of particle removal, for example microbial viability post-filtration is less frequently studied and a large portion of the available data was derived from a laboratory setting rather than real-world clinical environments. Also, it has been observed that the incubation temperatures after sampling significantly affected the results of microbial levels (Dai et al. 2021). Since most of the studies use different incubation temperatures and times, this can lead to varied distorted results. In addition, there are a lot of fake HEPA filters on the market, which have lower filtering efficiency when compared to authentic ones but still present a higher filtration efficiency in their marketing. Many studies failed to report the type of HEPA filter used, leaving suspicions of their authenticity. This could potentially dilute the true microbiological efficiency of unmodified authentic HEPA filtration. The visual presence of HEPA units could induce a false sense of security in humans, which can be a good notion. Although in a worst-case scenario, this could in turn lead to subconscious reductions in other preventive measures, which could serve negatively towards overall reductions (Falkenberg et al. 2023).

The study was limited by many of the references being secondary data and literature-based analysis, rather than clinical trials or in situ measurements. Future research should focus on quantifying the long-term microbiological performance of HEPA filters in operational healthcare environments, evaluating microbial colonization risk over time and assessing the safety and effectiveness of antimicrobial enhancements under clinical conditions. Thus, a well-designed and well-executed extensive, unifying, comprehensive and comparative study on the overall filtration efficiency and antimicrobial activity between different grades of filters (ULPA, HEPA, EPA) should be carried out, with both theoretical and experimental in situ data to back up the results. This study should also incorporate different antimicrobial technologies, evaluate their effectiveness with different filters and derive positive and negative conclusions regarding each filter technology combination. This way, it is possible to deduce the definite best, most cost-effective way to combat airborne microbial activity in medical facilities, which will have a huge impact on overall microbial activity in these facilities, leading to numerous consequent positive effects. This study would be expensive and time consuming, but in the long run the possible all-encompassing savings with such information would exceed the relatively short-lived expenses by tremendous amounts. This, combined with the preservation of human life, should easily justify such a study.

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